Shiv Nadar University - Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 28/09/2026 (Week 8) **Continuous Lab Evaluation-9** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Allotted Marks: 10 (Hardware Implementation + Code Efficiency + Viva-Voce)

Topics: CubeMX + Keil MDK + STM32 Development Board

STM32 HAL libraries are allowed

Supporting Documents: **1.** Board Schematics **2.** Reference Manual

STM32 UART Command Controlled LED

Problem Statement:

Task 9A — Basic UART Communication

Configure UART using STM32CubeMX and HAL, establish communication between the STM32 and a PC terminal, and implement character and string transmission/reception.

- **Transmit a Character** - Configure the STM32 UART for a standard serial configuration and transmit a character repeatedly to a PC terminal.
- **Receive and Echo** - Modify the code so that every character received from the PC is transmitted back to the PC.

Task 9B — String-Based Application Logic

Develop a UART-based command interface for an STM32 device. The system shall accept commands from a terminal and control the onboard LED. For example:

> LED ON	> LED OFF	> LED TOGGLE
LED turned ON	LED turned OFF	LED toggled

HAL APIs exclusively needed

For the basic UART experiment, try with first three APIs & Later explore others :

HAL API	Purpose
HAL_UART_Init()	**Initialize UART peripheral**
HAL_UART_Transmit()	**Transmit characters/string**
HAL_UART_Receive()	**Receive characters/string**
HAL_UART_Transmit_IT()	Interrupt-driven transmission — later
HAL_UART_Receive_IT()	Interrupt-driven reception — later
HAL_UART_Transmit_DMA()	DMA transmission — advanced
HAL_UART_Receive_DMA()	DMA reception — advanced

Students should investigate:	**Test 2:**	**Test 4:**
'\r' → Carriage Return	"HELLO\r"	"HELLO\r\nWORLD\r\n"
'\n' → Line Feed	**Test 3:**	
Create a small test program:	"HELLO\n"	Record what happens in PuTTY.
Test 1:		Then ask them to explain:
"HELLO\r\n"		**Why are both CR and LF**